

CONCEPT 30.4**Human welfare depends on seed plants (pp. 651–652)**

- Humans depend on seed plants for products such as food, wood, and many medicines.
- Destruction of habitat threatens the extinction of many plant species and the animal species they support.

? Explain why destroying the remaining tropical forests might harm humans and lead to a mass extinction.

TEST YOUR UNDERSTANDING

➔ For more multiple-choice questions, go to the **Practice Test** in the **Mastering Biology** eText or Study Area, or go to goo.gl/GruWRg.

Levels 1-2: Remembering/Understanding

- Where in an angiosperm would you find a megasporangium?
 - in the style of a flower
 - enclosed in the stigma of a flower
 - within an ovule contained within an ovary of a flower
 - packed into pollen sacs within the anthers found on a stamen
- Which of the following is a key feature of seed plants facilitating life on land?
 - homospory
 - pollen
 - reduced sporophytes
 - spores
- In angiosperms, which of the following is correctly paired with its chromosome count?

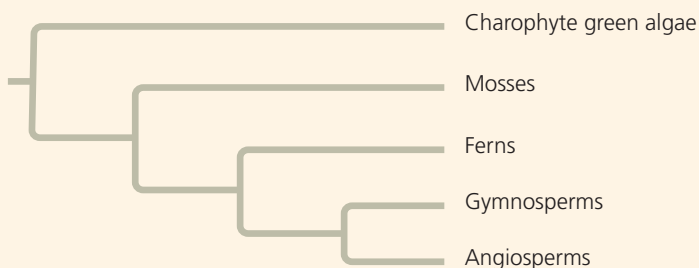
(A) microspore— n	(C) egg— $2n$
(B) zygote— n	(D) megaspore— $2n$
- Which of the following is a characteristic that distinguishes gymnosperms and angiosperms from other plants?
 - alternation of generations
 - independent gametophytes
 - vascular tissue
 - ovules
- Which of the following are present in angiosperms but not in gymnosperms?

(A) seeds	(C) ovaries
(B) pollen	(D) ovules

Levels 3-4: Applying/Analyzing

- DRAW IT** Use the letters a–d to label where on the phylogenetic tree each of the following derived characters appears.

a. flowers	c. seeds
b. embryos	d. vascular tissue



- EVOLUTION CONNECTION** The history of life has been punctuated by several mass extinctions. For example, the impact of a meteorite may have wiped out most of the dinosaurs and many forms of marine life at the end of the Cretaceous period (see Concept 25.4). Fossils indicate that plants were less severely affected by this mass extinction. What adaptations may have enabled plants to withstand this disaster better than animals?

Levels 5-6: Evaluating/Creating

- SCIENTIFIC INQUIRY • DRAW IT** As will be described in detail in Concept 38.1, the female gametophyte of angiosperms typically has seven cells, one of which, the central cell, contains two haploid nuclei. After double fertilization, the central cell develops into endosperm, which is triploid. Because magnoliids, monocots, and eudicots typically have female gametophytes with seven cells and triploid endosperm, scientists assumed that this was the ancestral state for angiosperms. Consider, however, the following recent discoveries:
 - Our understanding of angiosperm phylogeny has changed to that shown in Figure 30.14b.
 - *Amborella trichopoda* has eight-celled female gametophytes and triploid endosperm.
 - Water lilies and star anise have four-celled female gametophytes and diploid endosperm.
 - Draw a phylogeny of the angiosperms (see Figure 30.14b), incorporating the data given above about the number of cells in female gametophytes and the ploidy of the endosperm. Assume that all of the star anise relatives have four-celled female gametophytes and diploid endosperm.
 - What does your labeled phylogeny suggest about the evolution of the female gametophyte and endosperm in angiosperms?
- WRITE ABOUT A THEME: ORGANIZATION** Cells are the basic units of structure and function in all organisms. A key feature in the life cycle of plants is the alternation of multicellular haploid and diploid generations. Imagine a lineage of flowering plants in which mitotic cell division did not occur between the events of meiosis and fertilization (see Figure 30.12). In a short essay (100–150 words), describe how this change in the timing of cell division would affect the structure and life cycle of plants in this lineage.

10. SYNTHESIZE YOUR KNOWLEDGE

This photograph shows a milkweed seed in flight. Describe how seeds and other adaptations in seed plants contributed to the rise of seed plants and their dominant role in plant communities today.

For selected answers, see Appendix A.

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➔ **Instructors:** Questions can be assigned in Mastering Biology.